Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and product preferences from **3,900 transactions** to guide strategic decisions.

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Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating only

Demographics

Age, Gender, Location, Subscription

Purchase Details

Item, Category, Amount, Season

Behavior

Discount, Frequency, Rating, Shipping

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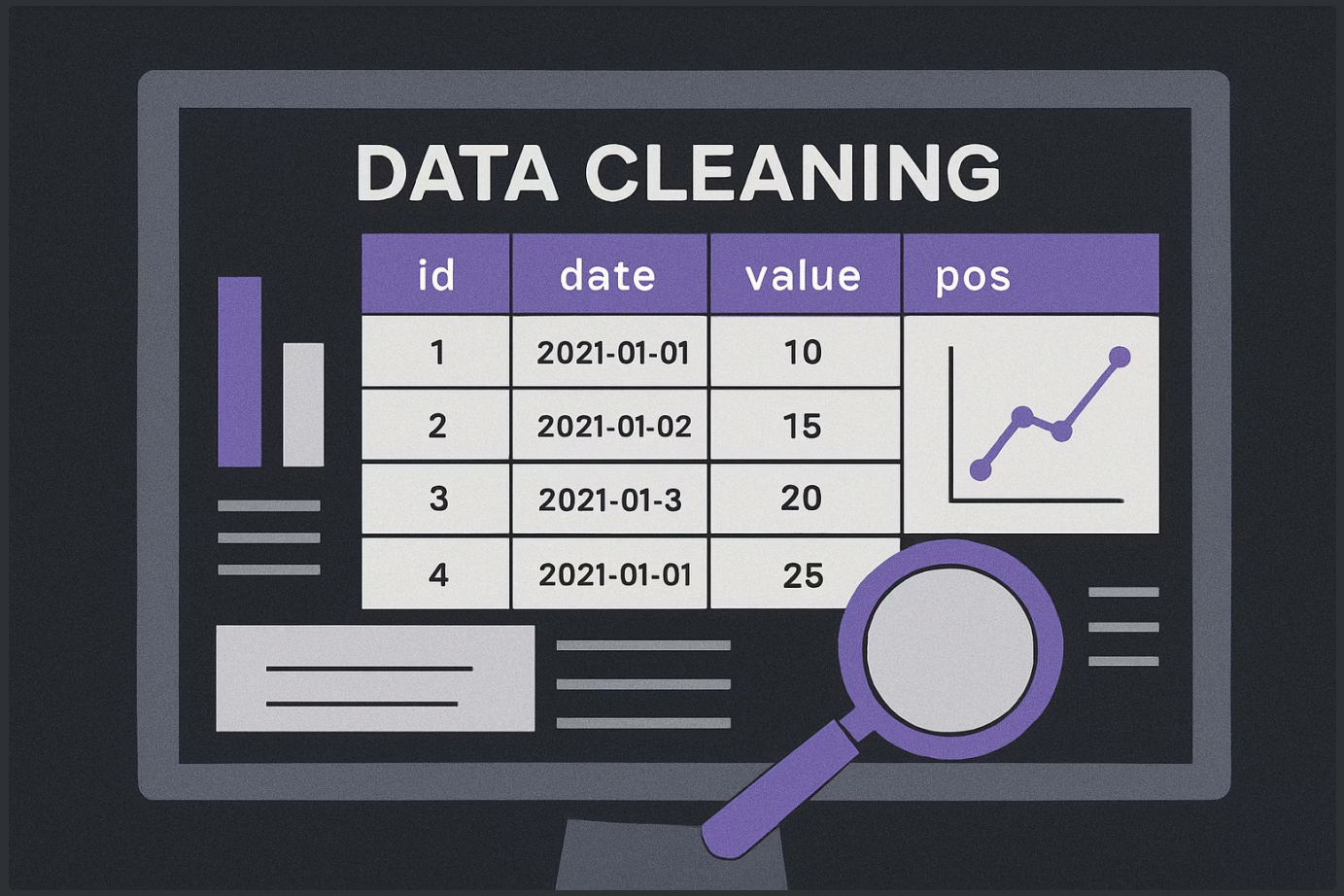
Python: Data Preparation



Load & Explore

`pandas` — `df.info()`, `.describe()`

Clean & Standardize

Imputed missing Review Ratings; renamed to snake_case

Feature Engineering

Created `age_group` & `purchase_frequency_days`

PostgreSQL Load

Cleaned DataFrame loaded for SQL analysis

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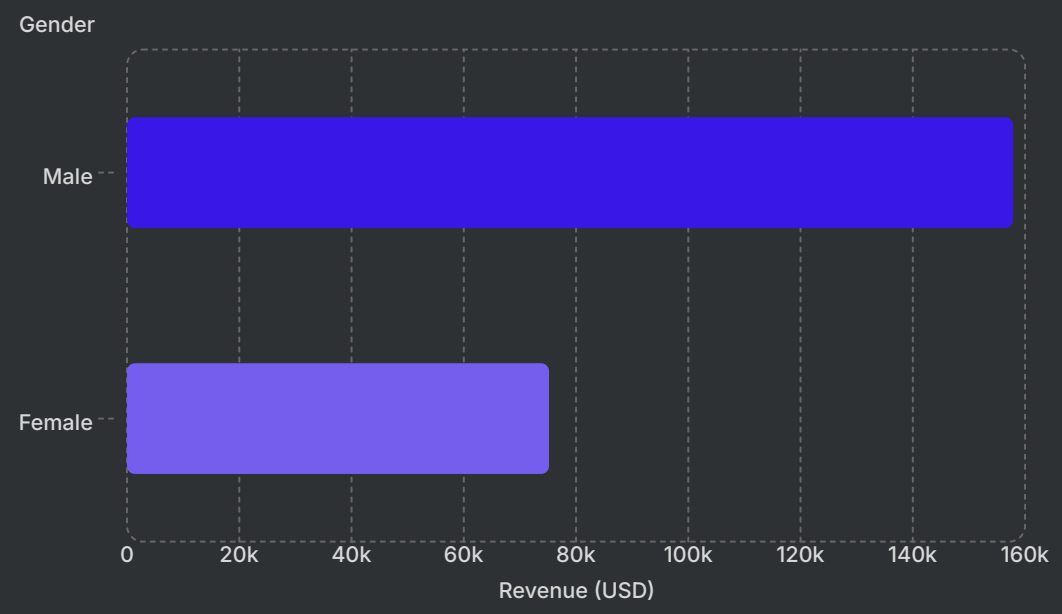
SQL Insights: Revenue & Shipping



Revenue by Gender

Male customers generated **2× more revenue** than female customers.

Shipping Type Avg. Spend

- Express: **$60.48**
- Standard: **$58.46**

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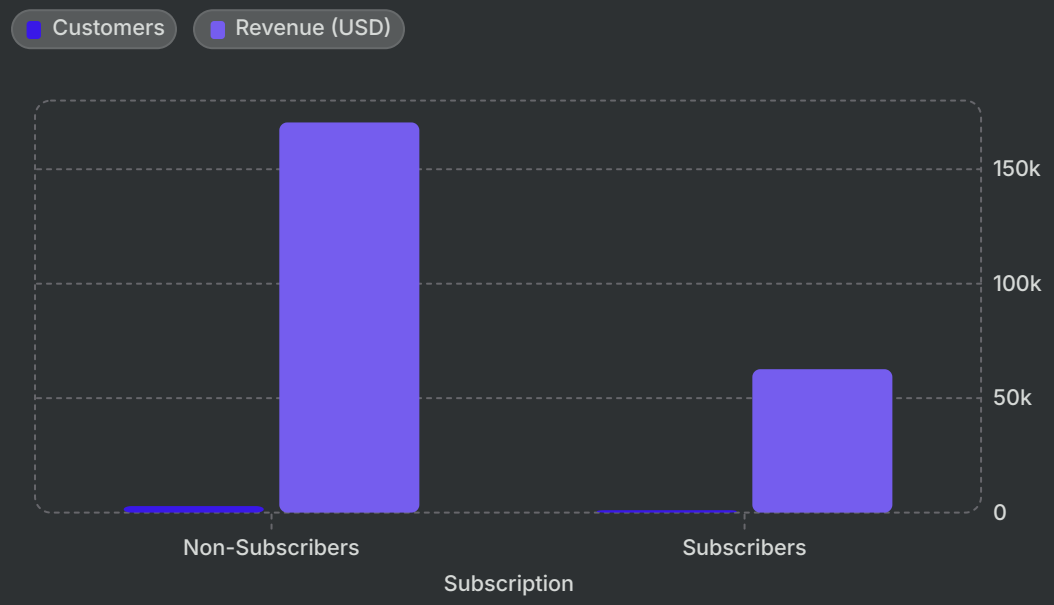
Subscriptions & Discounts



Subscription Gap

Only **27%** subscribed — non-subscribers drive **73%** of revenue.

Top Discount-Dependent Products

1. Hat — 50%
2. Sneakers — 49.66%
3. Coat — 49.07%
4. Sweater — 48.17%
5. Pants — 47.37%

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Customer Segmentation

🏆 Loyal

3,116 customers

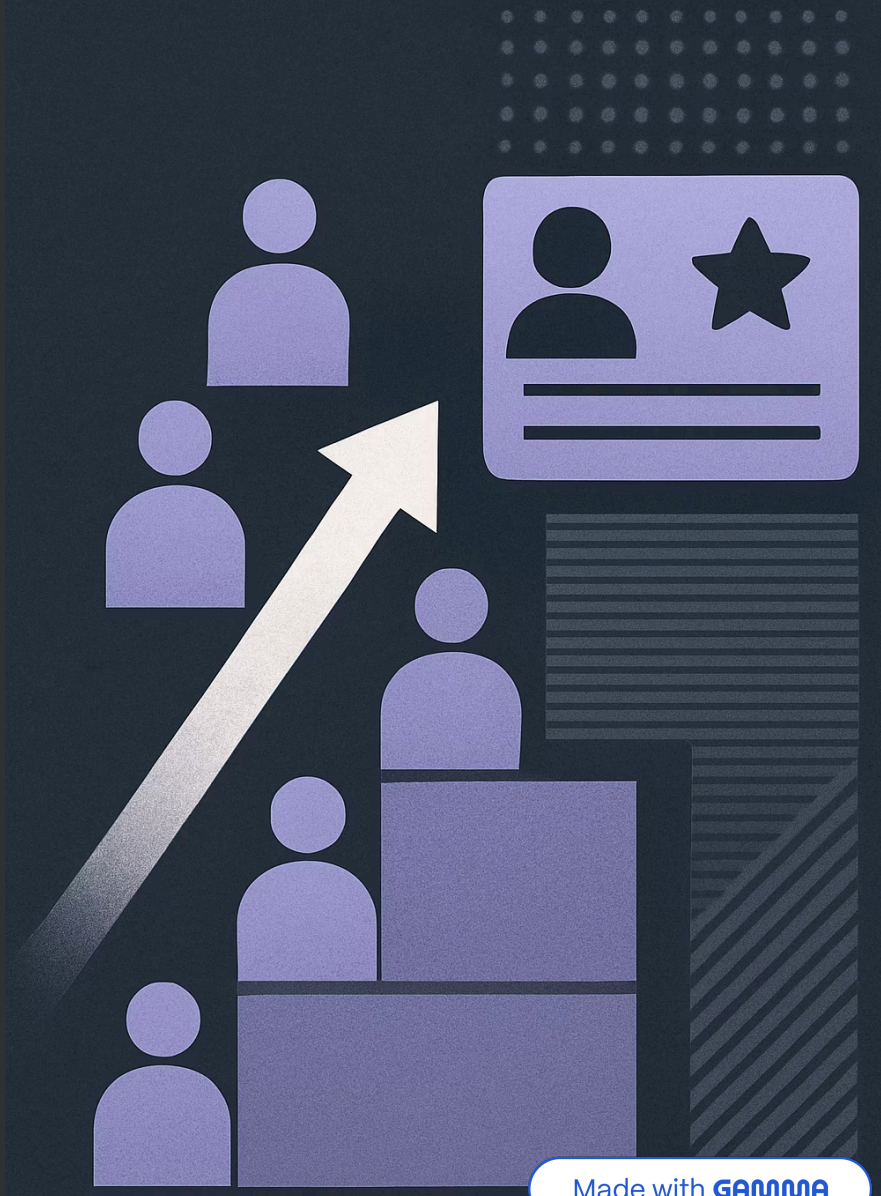
🔁 Returning

701 customers

🆕 New

83 customers

Repeat Buyers: 2,518 non-subscribers vs. 958 subscribers with 5+ purchases — subscription conversion opportunity.



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Power BI Dashboard

Key Metrics Visualizations

- Revenue & Sales by Category
- Revenue by Age Group
- % Customers by Subscription

$59.76	3.75	3.9K
Avg. Purchase	Avg. Rating	Customers

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers toward "Loyal" segment

Review Discount Policy

Balance sales lift with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

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